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NPS/003/019 – Technical Specification for Electrical Insulating Fluids for use in Plant & Switchgear.

1. Purpose

This specification details the technical requirements for electrical insulating oil for use in plant and switchgear on the CE Electric UK electrical distribution networks.

This document supersedes the following document, all copies of which shall be destroyed.

Ref	Version	Title
NPS/003/019	3.0	Technical Specification for Electrical Insulating Fluids for use in Plant and Switchgear

2. Scope

This specification covers new and reclaimed electrical insulating oil, and other fluids, for use in ground mounted and pole mounted, plant and switchgear on the CE Electric UK electrical distribution networks

This specification excludes insulating fluids for use in cables and insulating fluids for use in capacitors.

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3. Technical Requirements

3.1 Electrical Insulating and Cooling Fluid.

New or reclaimed electrical insulating oil shall be used. It shall be: Naphthenic, uninhibited, unpassivated, mineral oil complying fully with the requirements of sections 3.1.1 and 3.1.2. In either case the conditions regarding corrosive sulphur content shall apply as described in section 3.1.3.

3.1.1 Unused oil

Oil shall be naphthenic, uninhibited, unpassivated mineral oil of classification *Transformer Oil*, U, -30°C in compliance with the requirements of BS EN 60296: 2004: *Fluids for electrotechnical application: Unused mineral insulating oils for transformers and switchgear*, with the additional requirement of:

- A gassing tendency limited to less than 5mm³/min at 50 Hz after 120 minutes when tested in accordance with test method A of BS5797:1986

3.1.2 Reclaimed Oil

Oil shall be Naphthenic, uninhibited, unpassivated mineral oil complying fully with the requirements of BS148:2009: *Reclaimed mineral insulating oil for transformers and switchgear - specification*, with the additional requirement of:

- A gassing tendency of less than 5mm³/min when tested in accordance with test method A of BS5797:1986
- PCB content shall be minimised as far as practicable, but in any case shall be less than 10ppm

3.1.3 Corrosive Sulphur.

In addition to the requirements of BS148 and IEC 60296 the following shall apply

- Corrosive sulphur and potentially corrosive sulphur content shall be classed as 'Non-corrosive' as determined by the test methods prescribed in IEC 62535:2008.

And

- Corrosive sulphur and potentially corrosive sulphur content shall be classed as 'Non-corrosive' as determined by the test methods prescribed in ASTM D 1275B (E 2005).

3.2 Fluids for Use in Non-standard Situations.

Unused or reclaimed electrical insulating oil as specified in 3.1 above shall be used for standard applications.

Other fluids will be considered for special applications; where there is unacceptable fire risk, environmental sensitivity or environmental risk. Other fluids may include synthetic oil, or vegetable based esters in accordance with or generally in accordance with IEC 61099 and/or IEC 60836; for example Midel 7131, Envirotemp FR3, Dow Corning 561, etc.

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Before use on the CE Electric UK distribution networks these fluids must be assessed and formally approved by CE Electric UK's Standards and Assessment Section.

3.3 Amendments from Previous Version

Section	Subject	Amendments
1.	Purpose	References to NEDL and YEDL replaced with CE Electric UK
2.1	Contents	Section Added
3	Technical Requirements	The specification has been amended to reflect changes to BS148, but the products defined remain the same; i.e. new or reclaimed, uninhibited, unpassivated mineral oil with a minimum energisation temperature of -30oC. <u>Reclaimed oil</u> Reference is made to BS 148: 2009. This BS is now restricted to reclaimed oil whereas BS 148: 1998 covered both unused and reclaimed oils. Classes I, II and III are no longer contained in BS148: 2009 but the standard corresponds closely to the old Class I, previously specified by CE Electric. <u>New (unused) oil</u> Reference to BS 148 is removed. Reference is made to BS EN 60296: 2004 not IEC 60296: 2003, but these two documents are identical in content.
3.3	Amendments From Previous Version	Section Added

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4. References

4.1 External Documentation

Reference	Title
BS148:2009	Reclaimed mineral insulating oil for transformers and switchgear - Specification
BS EN 60296:2004	Fluids for electrotechnical applications - Unused mineral insulating oils for transformers and switchgear
ASTM D 1275B (E 2005)	Standard Test Method for Corrosive Sulfur in Electrical Insulating Oils - IP 315/98
IEC 62535:2008	Test Method for Corrosive Sulphur in Electrical Insulating Oils
IEC 61099:1992	Specifications for Unused Synthetic Organic Esters for Electrical Purposes - First Edition
IEC 60836:2005	Specifications for unused silicone insulating liquids for electrotechnical purposes - Edition 2

4.2 Internal documentation

Reference	Title
NONE	

5. Definitions

Term	Definition
NONE	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
K Midgley	Plant & Switchgear Standards Manager	K Midgley	02/11/2009

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	04/11/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D Blackledge	Plant and Switchgear Standards Manager	D Blackledge	04/11/2009

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	16/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/11/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	05/11/2009

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Appendix 1 – Reclaimed Oil: Declaration of Conformance to BS148:2009 and to NPS/003/019

BS148:2009 Clause	Requirement	Conformance and Details
3.1 - additive	List any additives and quantities/proportions	
3.4 – passivator	Unpassivated	
3.10 – uninhibited oil	Uninhibited	
4.1 Tanker wagons and drums	List vessels used and detail cleaning procedures for each	
4.2 – Marking of vessels	List vessels used and detail vessel marking for each: Minimum a and b markings per BS148 2009	
4.3 – Accompanying documentation	Provide samples of documentation: Minimum a, b, c statements per BS148 2009	
4.4 - Health & Safety information	Provide COSHH data sheets, etc.	
5 - Sampling	Top AND bottom samples shall be taken (procedure per BS5263) and analysed and results supplied to CEEUK.	
6 – Properties of the oil	Oil shall be additive-free uninhibited, unpassivated and within limiting values per Table 1 of BS148 2009	
Appearance	Clear, free from sediment and suspended matter.	
Density at 20°C	$\leq 0.895\text{kg/dm}^3$	
Kinematic viscosity 40°C	$\leq 13\text{mm}^2/\text{s}$	
- 15°C	$\leq 800\text{mm}^2/\text{s}$	
Flash point	$\geq 135^\circ\text{C}$	
Pour point	$\leq -30^\circ\text{C}$	
Neutralisation value	$\leq 0.03\text{ mg KOH per g of oil}$	
Corrosive sulphur	Non-corrosive per IEC 62535:2008 and Non-corrosive per ASTM D 1275B (E 2005)	
Passivator content	Not detectable	

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BS148:2009 Clause	Requirement	Conformance and Details
Water content Bulk delivery Drum delivery	<= 20mg/kg <= 30mg/kg	
Anti-oxidant additives	Not Detectable	
Oxidation stability 164 hr	Total acidity <= 1.2mg KOH per g of oil Sludge <= 0.8% by mass	
Breakdown voltage as delivered	>=30kV	
Dielectric dissipation factor	<= 0.005 at 90°C and 40 Hz to 62 Hz	
Total PCB content	< 10mg/kg	
Total furans	< 0.2 mg/kg	
Gassing Tendency at 50 Hz after 120 min., Method A	<= 5 mm ³ /min	

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Appendix 2 – Unused Oil: Declaration of Conformance to IEC60296:2004 and to NPS/003/019

IEC60296 Clause	Requirement	Conformance and Details
1 - Unused oil	Confirm oil is 'Unused oil'	
3.3 - Additives	CEEUK require additive free oil. List any additives and quantities/proportions	
4.4 - Health & Safety information	Provide COSHH data sheets, etc.	
5.1.1 Class	Transformer oil	
5.1.2 - Inhibitor content	Uninhibited: U	
5.1.3 – Lowest cold start energising temperature (LCSET)	-30°C	
5.4 – Identification and delivery		
a) vessels clean and suitable	a) List vessels used and detail cleaning procedures for each	
b) vessel marking	b) List vessels used and detail vessel marking for each	
c) accompanying documentation	c) Provide samples of documentation:	
5.5 - Sampling	Top AND bottom samples shall be taken (procedure per IEC 60475) and analysed and results supplied to CEEUK.	
6 – Properties of the oil	Oil shall be additive-free uninhibited, unpassivated and within limiting values per Tables 1 and 2 BS EN 60296: 2004	
Viscosity at 40 °C	Max. 12 mm ² /s	
Viscosity at –30 °C	Max. 1 800 mm ² /s	
Pour point	Max. –40 °C	
Water content	=< 30 mg/kg (bulk supply) =< 40 mg/kg (drums)	

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IEC60296 Clause	Requirement	Conformance and Details
Breakdown voltage	Min. 30 kV / 70 kV	
Density at 20 °C	Max. 0,895 g/ml	
DDF at 90°C	Max. 0,005	
Appearance	Clear, free from sediment and suspended matter	
Acidity	Max. 0,01 mgKOH/g	
Interfacial tension	Min. 40mN/m	
Total sulphur content	No general requirement	
Corrosive sulphur	Not corrosive Plus: Non-corrosive per IEC 62535:2008 and: Non-corrosive per ASTM D 1275B (E 2005)	
Antioxidant additive	(U) uninhibited oil: not detectable	
2-Furfural content	Max. 0,1 mg/kg	
Oxidation stability	(U) Uninhibited oil: 164 h	
Total acidity	Max. 1,2 mg KOH/g	
Sludge	Max. 0,8 %	
DDF at 90 °C	Max. 0,500	
Gassing	<= +5mm ³ /min at 50 Hz after 120 minutes	
Flash point	Min. 135 °C	
PCA content	Max. 3 %	
PCB content	Not detectable	